

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

***CO1:** Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks: 25

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I S.Dharani Teja certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. Dharaniteja

4.Examine the training and validation loss over multiple epochs.

Epoch	Training Loss	Validation Loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9

Step 1: Observe the Training Loss

- Epoch 1 → 2.8
- Epoch 2 → 2.5 ↓
- Epoch 3 → 2.2 ↓
- Epoch 4 → 2.0 ↓
- Epoch 5 → 1.8 ↓
- Epoch 6 → 1.6 ↓
- Epoch 7 → 1.5 ↓
- Epoch 8 → 1.3 ↓

Observation: Training loss continuously decreases.

Step 2: Observe the Validation Loss

- Epoch 1 → 2.4
- Epoch 2 → 2.2 ↓
- Epoch 3 → 2.0 ↓ (Minimum)
- Epoch 4 → 2.1 ↑
- Epoch 5 → 2.3 ↑
- Epoch 6 → 2.6 ↑
- Epoch 7 → 2.9 ↑
- Epoch 8 → 3.2 ↑

Observation: Validation loss starts increasing after Epoch 3.

Step 3: Identify Overfitting

Training loss ↓ while Validation loss ↑

Therefore,

Overfitting begins at Epoch 4.

Step 4: Two Techniques to Reduce Overfitting

Technique 1: Early Stopping

- Stop training when validation loss increases.

Technique 2: Dropout

- Randomly disables neurons during training.
- Prevents memorization.

Step 5: Effect of Overfitting

- Training performance improves.
- Testing performance decreases.
- Model memorizes data.
- Poor generalization on unseen data.

6 . Generalization and Regularization

Epoch	Training Loss	Validation Loss
1	3.0	2.7
2	2.6	2.3
3	2.3	2.1
4	2.1	2.0
5	1.9	2.2
6	1.7	2.5
7	1.6	2.8

Step 1: Observe Training Loss

3.0 → 2.6 → 2.3 → 2.1 → 1.9 → 1.7 → 1.6 → 1.4

Training loss decreases continuously.

Step 2: Observe Validation Loss

2.7 → 2.3 → 2.1 → **2.0 (Lowest)** → 2.2 → 2.5 → 2.8 → 3.1

Validation loss is lowest at **Epoch 4**.

Step 3: Best Generalization

The model performs best when validation loss is minimum.

Best Generalization = Epoch 4

Step 4: Identify Overfitting

After Epoch 4:

Training loss decreases.

Validation loss increases.

Therefore,

Overfitting starts after Epoch 4 (from Epoch 5 onwards).

Step 5: Regularization Techniques

1. L2 Regularization

- Penalizes large weights.
- Prevents overfitting.

2. Dropout

- Randomly removes neurons during training.
- Improves model generalization.

8. Performance Metrics

Given

TP = 138

FN = 12

TN = 135

$$FP = 15$$

Step 1: Total Samples

Formula

$$\text{Total} = TP + TN + FP + FN$$

Substitute

$$= 138 + 135 + 15 + 12 = 138 + 135 + 15 + 12 = 138 + 135 + 15 + 12 = 300 = 300 = 300$$

$$\text{Total} = 300$$

Step 2: Accuracy

Formula

$$\text{Accuracy} = (TP + TN) / \text{Total} * 100$$

Substitute

$$= (138 + 135) / 300 * 100$$

$$= 273 / 300 * 100$$

$$= 91\%$$

Step 3: Precision

Formula

$$\text{Precision} = TP / (TP + FP) * 100$$

Substitute

$$= 138 / (138 + 15) * 100$$

$$= 138 / 153 * 100$$

$$= 90.20\%$$

Step 4: Recall (Sensitivity)

Formula

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) * 100$$

Substitute

$$= 138 / (138 + 12) * 100$$

$$= 138 / 150 * 100$$

$$= 92\%$$

Step 5: Specificity

Formula

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP}) * 100$$

Substitute

$$= 135 / (135 + 15) * 100$$

$$= 135 / 150 * 100$$

$$= 90\%$$

Step 6: F1-Score

Formula

$$\text{F1} = 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall})$$

Substitute

$$= 2 * 90.20 * 92 / (90.20 + 92)$$

$$= 16596.8 / 182.2$$

$$= 91.1\%$$

Final Answer

$$\text{Accuracy} = \mathbf{91\%}$$

$$\text{Precision} = \mathbf{90.20\%}$$

$$\text{Recall} = \mathbf{92\%}$$

$$\text{Specificity} = \mathbf{90\%}$$

$$\text{F1-Score} = \mathbf{91.1\%}$$

15 . Loss Analysis

Epoch	Training Loss	Validation Loss
1	2.8	2.6
2	2.4	2.2
3	2.1	1.9
4	1.9	1.8
5	1.7	1.9
6	1.5	2.1
7	1.3	2.4

Step 1: Observe Training Loss

2.8 → 2.4 → 2.1 → 1.9 → 1.7 → 1.5 → 1.3 → 1.2

Training loss continuously decreases.

Step 2: Observe Validation Loss

2.6 → 2.2 → 1.9 → **1.8 (Minimum)** → 1.9 → 2.1 → 2.4 → 2.7

Validation loss is minimum at **Epoch 4**.

Step 3: Best Generalization

The model gives its best performance when validation loss is lowest.

Best Generalization = Epoch 4

Step 4: Identify Overfitting

After Epoch 4:

- Training loss keeps decreasing.
- Validation loss increases.

Therefore,

Overfitting begins after Epoch 4.

Step 5: Two Techniques to Reduce Overfitting

1. Early Stopping

- Stops training once validation loss starts increasing.

2. Dropout

- Randomly deactivates neurons during training.
- Prevents memorization and improves generalization.

Question 10 .Quality Inspection System

Given

True Positive (TP) = 230 (Defective identified as Defective)

False Negative (FN) = 20 (Defective identified as Non-Defective)

True Negative (TN) = 225 (Non-Defective identified as Non-Defective)

False Positive (FP) = 25 (Non-Defective identified as Defective)

Step 1: Draw the Confusion Matrix

Actual / Predicted	Defective	Non-Defective
Defective	TP = 230	FN = 20
Non-Defective	FP = 25	TN = 225

Step 2: Calculate Total Number of Products

Formula

$$\text{Total} = \text{TP} + \text{TN} + \text{FP} + \text{FN}$$

Substitute the values

$$= 230 + 225 + 25 + 20$$

$$\text{Total Products} = 500$$

Step 3: Calculate Accuracy

Formula

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} * 100$$

Substitute

$$= 230+225/500 * 100$$

$$= 455/500 * 100$$

$$= 91 \%$$

Accuracy = 91%

Step 4: Calculate Precision

Formula

$$\text{Precision} = TP/TP+FP*100$$

Substitute

$$=230/230+25*100$$

$$=230/255 *100$$

$$= 90.20\%$$

Precision=90.20%

Step 5: Calculate Sensitivity (Recall)

Formula

$$\text{Recall} = TP/TP+FN*100$$

Substitute

$$= 230/230+20 *100$$

$$=230/250 *100$$

$$=92\%$$

Sensitivity(Recall) = 92%

Step 6: Calculate Specificity

Formula

$$\text{Specificity} = TN/TN+FP*100$$

Substitute

$$=225/225+25 *100$$

$$= 225/250 * 100$$

$$= 90\%$$

Specificity = 90%

Step 7: Calculate F1-Score

Formula

$$\mathbf{F1} = 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall})$$

Substitute

$$= 2 * 90.92 * 92 / (90.92 + 92)$$

$$= 16596.8 / 182.92$$

$$= 91.1\%$$

F1-Score = 91.1%

Interpretation

The quality inspection model achieves **91% accuracy**, **90.20% precision**, **92% sensitivity**, **90% specificity**, and an **F1-score of 91.1%**.